

Evil Dares

An exploration using the Hypergeometric Probability
distribution

Mathematics Higher Level

Internal Assessment

Candidate Code

gny997

Table of contents

1. Introduction	1
2. What is the hypergeometric distribution?	1
3. The probability mass function	3
4. Visual Representation	5
5. The expected value of the hypergeometric distribution	6
6. The variance of the hypergeometric distribution	8
7. Solving the opening problem	11
8. Conclusion	13

1. Introduction

Before beginning this IA, I would like to tell a story of something which occurred to me a few weeks ago. I was at a house party and someone suggested a game in which I instantly saw a large quantity of mathematics at play. The game had the following rules:

1. A box is prepared that has a number of things written on slips of paper that people have to do, a certain amount are “good” ones (blank) and a certain amount are “evil” ones. The version we played had 50 slips, 25 of each. Evil dares included, e.g. eating a spoon of flour or cinnamon or to jump into the ice cold pool.
2. People go around in circles and draw a specific amount of slips from the box. We played so that everyone had to pick 7 slips.
3. If a player has 3 or less evil slips, he or she has to do all of them.
4. If a player has 4 or more evil slips, then they only have to do one of them.

What followed this is that I wanted to find a way to calculate the probability of having “lost”, meaning having to do all three “evil” dares. I wanted to find the probability of this happening with the rules in this game, but researching this then also made me interested in how changing certain parameters would also change the outcome, so the probability of said certain outcome taking place. Once I researched a little more into how this probability could be calculated, I came across the hypergeometric probability distribution. We had briefly mentioned it in class, when I found the feature for it on my GDC, but we did not delve deeper. I noticed that the hypergeometric distribution and the conditions around it follows very closely the game that we played at the party, and thus, I decided to investigate it further and the properties it has, to ultimately find the probability of having “lost”. I wanted to also *derive* formulas for the expected value and variance as when asking my math teacher, he said it would be too complicated, and I wanted to prove him wrong. I will also be solving the opening problem and speaking about some of the real life limitations of such a model.

2. What is the hypergeometric distribution?

The hypergeometric distribution is a probability distribution that is not part of the IB Maths HL syllabus. It is a discrete probability distribution, meaning that the input parameters and the variable are of a discrete nature, and can only take natural number values. For now, we can say that the hypergeometric distribution is used to determine the probability of r number of successes in a sample of n trials, from a larger population of size N with a total

of R successes.¹ We will look at the more complex example later, but in order to understand the concept, I will first conduct this basic example:
 Let us say we have a box containing a total of 64 coloured marbles, 28 of which are red, and 36 of which are blue.

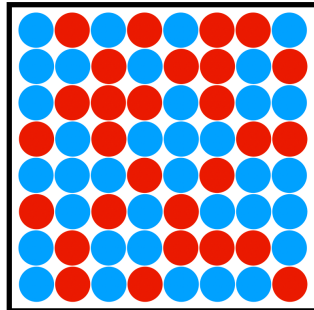


Fig. 1: Marbles in a box (own creation, in Pages)

We are assuming that the marbles are randomly distributed throughout the box. Now let us take a handful of marbles, say 20, out of the box.

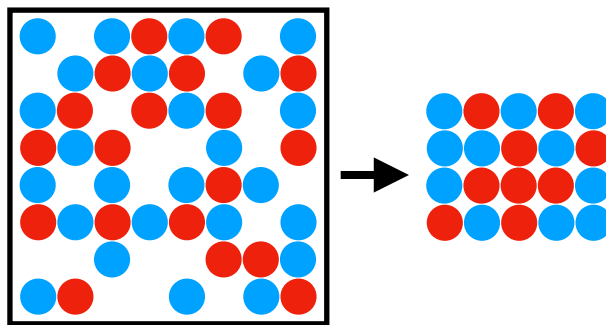


Fig. 2: Sample of marbles taken from box (own creation, in Pages)

We can now calculate the probability that this handful (sample) will contain a certain amount of successes, or red marbles. The distribution of these probabilities against values of r is called a *hypergeometric distribution*. Combinations are used since the order of the sample doesn't matter. The formula for the probability that X takes a certain value of r is given as:

$$P(X = r) = \frac{\overbrace{\binom{R}{r}}^{\text{Combinations of successes in the sample}} \underbrace{\binom{N-R}{n-r}}_{\text{Combinations of failures in the sample}}}{\binom{N}{n}} \quad (1)$$

So for our little example with the marbles, let us define the variables. Since our large box has 64 marbles, $N = 64$. We have 28 red marbles, so $R = 28$. Finally, our sample size is 20, so $n = 20$. Now our probability mass function looks like this:

¹ "Hypergeometric Distribution." *Wolfram MathWorld*, Wolfram Research, mathworld.wolfram.com/HypergeometricDistribution.html.

$$P(X = r) = \frac{\binom{28}{r} \binom{36}{20-r}}{\binom{64}{20}} \quad (2)$$

Eq. 2: Probability Mass function with parameters substituted

So as an example, let us calculate the probability that there will be 14 red marbles in the sample. For this, we need to substitute $r = 14$ in the function, which gives us this:

$$\begin{array}{ccc} \text{Combinations of} & & \text{Combinations} \\ \text{successes in the sample} & P(X = 14) = \frac{\binom{28}{14} \binom{36}{6}}{\binom{64}{20}} & \text{of failures} \end{array} \quad (3)$$

Evaluating combinatorial expressions:

$$P(X = 14) = \frac{\frac{28!}{14!14!} \frac{36!}{6!30!}}{\frac{64!}{20!44!}} \quad (4)$$

$$P(X = 14) = \frac{40116600 \cdot 1947792}{19619725782651120}$$

$$P(X = 14) \approx 0.00398$$

To sum up, we can say that when drawing a sample of **20 marbles** out of a box that contains **64 marbles** with **28 red marbles**, the probability that this sample then contains **14 red marbles** is around 0.00398. Later on, we will also be looking into a way to provide a visual representation of this. In the next section, we will be going over why this probability mass function (eq. 1) looks the way it does and what exactly is being calculated.

3. The probability mass function

If a random variable X is distributed hypergeometrically, we can denote it like:

$$X \sim \text{Hypergeometric}(N, R, n)$$

However, for simplicity's sake, in this IA, we will be denoting it as:

$$X \sim HG(N, R, n)$$

For completion, the probability mass function of the hypergeometric distribution is:

$$P(X = r) = \frac{\binom{R}{r} \binom{N-R}{n-r}}{\binom{N}{n}} \quad (1)$$

Where:

- **X** is the random variable
- **R** is the amount of successes in the population
- **N** is the population size
- **n** is the sample size
- **r** is the amount of successes in the sample

Let us split up the probability mass function, in order to explain why it looks the way it does. The first term in the function is $\binom{R}{r}$. Since R is the total amount of successes in our population and r is the amount of successes in the sample, $\binom{R}{r}$ is the number of ways r successes can be selected out of a total population of R successes.

The next term in the function is $\binom{N-R}{n-r}$. Essentially, it is the “inverse” of the first term.

N - R is simply the amount of failures in the population, as N is the size of the population, and R is the amount of successes. Consequently, n - r is the amount of failures in the sample that we take. Hence, $\binom{N-R}{n-r}$ is simply the amount of ways that n - r failures can be selected out of a population of N - R failures.

It is important to notice that if we set the two terms next to each other, say:

$$\binom{R}{r} \text{ and } \binom{N-R}{n-r}$$

we can observe that the sum of the total amount of trials for each term, so $R + (N - R)$ is just N, and the same happens for the number of choices: $r + (n - r)$ is just n. This is important and beneficial as we then essentially leave no member of the population unaccounted for.

If we multiply:

$$\binom{R}{r} \cdot \binom{N-R}{n-r}$$

the result will be the *total amount of possibilities* of selecting n members out of a larger population with N members, given a parameter r , the amount of successes out of these n members in the sample. Now, to find the probability of this taking place, we need to divide by the total amount of possibilities of how we can pick a sample of n out of a population of N members.

This total amount of possibilities can also be found using combinations, which would be

$\binom{N}{n}$. Since we are dividing by this, the final function will look like this:

$$P(X = r) = \frac{\binom{R}{r} \binom{N-R}{n-r}}{\binom{N}{n}} \quad (1)$$

4. Visual Representation

Since a hypergeometric probability distribution is not continuous, we cannot graph a continuous graph, however we can create a bar graph that shows us the values of $P(X=r)$ for any value of r . For the example with the marbles, the random variable is distributed as such:

$$X \sim HG(64,28,20)$$

A graph for distribution can be seen in fig.3:

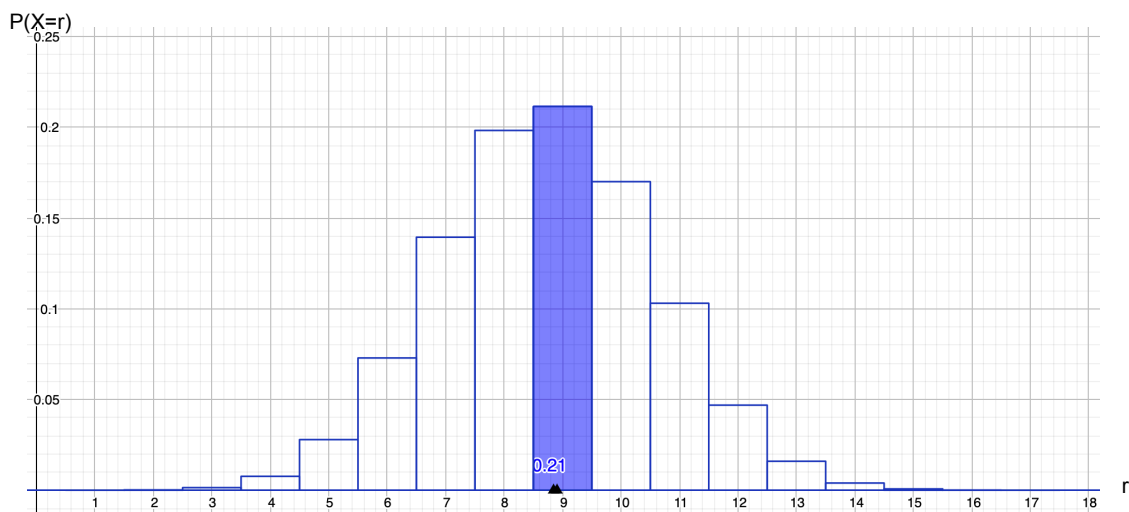


Figure 3: $X \sim HG(64,28,20)$ Visualised (Own graph, created using GeoGebra)

This should give a little more insight into how this example looks from a mathematical angle. Firstly, the height of the bar at $r = 14$ is around 0.00398, which suggests that our calculations were done correctly. Secondly, something else worth noticing is that the highest probability, or the mode/mean, takes place at $r = 9$, with a probability of ≈ 0.212 . It is important to see that the distribution is not symmetrical, with bias to one or the other side of the mode/mean. The mode/mean is easy to find, as it is the value of r with the highest probability. Now let us find two more characteristics that are a little more complicated to find, the expected value $E(X)$ and the variance $Var(X)$.

5. The expected value of the hypergeometric distribution

The expected value is the average value of r that we get out of all possible values for r .² In the example with the marbles, r was the amount of successes that occurred in a sample of 20. From this, we can deduce that $r \in \mathbb{N}, 0 \leq r \leq 20$.

We know that the expected value for any discrete random variable X is given as:

$$E(X) = \sum rP(X = r) \quad (5)$$

Meaning that we sum up the products of all values of r and their respective probabilities. In terms of our formula for hypergeometric distribution, this would look like eq. 6.

$$E(X) = \sum_{r=1}^n \frac{r \cdot \binom{R}{r} \binom{N-R}{n-r}}{\binom{N}{n}} \quad (6)$$

There are a couple of steps needed in order to get to a reasonable place. The first involves a relationship that I will quickly prove:

$$r \binom{R}{r} = R \binom{R-1}{r-1} \quad (7)$$

$$\begin{aligned} R \binom{R-1}{r-1} &= R \frac{(R-1)!}{(r-1)!(R-1-(r-1))!} \\ &= \frac{R!}{(r-1)!(R-r)!} \end{aligned}$$

² "Expected Value." *Stat Trek*, stattrek.com/statistics/dictionary.aspx?definition=expected%2Bvalue.

$$\begin{aligned}
R \binom{R-1}{r-1} &= \frac{r}{r} \cdot \frac{R!}{(r-1)!(R-r)!} \\
&= r \cdot \frac{R!}{r!(R-r)!} = r \binom{R}{r}
\end{aligned}$$

Now we can work with the function (eq. 6). Substituting $r \binom{R}{r} = R \binom{R-1}{r-1}$:

$$E(X) = \sum_{r=1}^n \frac{R \binom{R-1}{r-1} \binom{N-R}{n-r}}{\binom{N}{n}}$$

Any terms which don't involve r remain constant in the sum, meaning we can write them as a factor. This gives us:

$$E(X) = \frac{R}{\binom{N}{n}} \sum_{r=1}^n \binom{R-1}{r-1} \binom{N-R}{n-r}$$

We can substitute the value of $r - 1$ with another variable m to be able to move further. This would also change certain parameters in the sum. Simultaneously we are also trying to recreate the structure of the probability distribution function³. Also, since $m = r - 1$, $r = m + 1$:

$$E(X) = \frac{R}{\binom{N}{n}} \sum_{m=0}^{n-1} \binom{R-1}{m} \binom{N-1-(R-1)}{n-1-m}$$

Our goal is to eventually get the sum to equal 1. However, to do this, we need to have a complete probability mass function in the sum. For perspective, the parameters in the function will not be N , R , and n , but $N - 1$, $R - 1$, and $n - 1$. This means we still need a $\binom{N-1}{n-1}$ in the denominator. We can keep the function equivalent if we divide and multiply by a factor at the same time, hence:

³ Shahid, Iqbal. "Hypergeometric Distribution - Expected Value." *YouTube*, Google, Inc., 16 Apr. 2017, www.youtube.com/watch?v=MsoqHeThMGk.

$$E(X) = \frac{R \cdot \binom{N-1}{n-1}}{\binom{N}{n}} \sum_{m=0}^{n-1} \frac{\binom{R-1}{m} \binom{(N-1)-(R-1)}{(n-1)-m}}{\binom{N-1}{n-1}} \quad (8)$$

Upon closer inspection, the sum part of the function is a perfectly acceptable hypergeometric probability distribution function. Since we are summing all the probabilities in the given interval, all of this simplifies to 1. This leaves us with:

$$E(X) = \frac{R \cdot \binom{N-1}{n-1}}{\binom{N}{n}}$$

Now expanding:

$$E(X) = \frac{R \cdot \frac{(N-1)!}{(n-1)!(N-n)!}}{\frac{N!}{n!(N-n)!}}$$

Which finally leads us to:

$$E(X) = \frac{R \cdot \frac{(N-1)!}{(n-1)!}}{\frac{N!}{n!}} = R \cdot \frac{1}{\frac{\frac{N!}{n!}}{\frac{(N-1)!}{(n-1)!}}} = R \cdot \frac{1}{\frac{N}{n}} = R \cdot \frac{n}{N} \quad (9)$$

So now we know that for any random variable X distributed hypergeometrically, the expected value $E(X) = \frac{Rn}{N}$. For our example with the marbles, this lets the expected

value be:

$$E(X) = \frac{28 \cdot 20}{64} = 8.75$$

This is also the value that we get from GeoGebra when we insert the values into the probability calculator. This expected value is the mean/average of our distribution for these parameters. In a real life, we would round this to 9, but in theoretical sense, to calculate further, we would use 8.75.

6. The variance of the hypergeometric distribution

Now, we can also try to find an easy formula for the variance of a hypergeometric distribution, so how much it varies (is spread). We know that the variance of any discrete random variable X is:

$$Var(X) = \frac{\sum_{i=1}^k (x_i - \mu)^2}{n} \quad (10)$$

Essentially, what we are doing is finding the average $(X - \mu)^2$. Since the average is the expected value, this means we can also denote it as such⁴:

$$Var(X) = E((X - \mu)^2)$$

Now we can apply simple expectation algebra:

$$Var(X) = E((X - E(X))^2)$$

$$Var(X) = E(X^2 - 2XE(X) + E(X)^2)$$

$$Var(X) = E(X^2) - 2E(X)E(E(X)) + E(X)^2$$

$$Var(X) = E(X^2) - 2E(X)^2 + E(X)^2$$

$$Var(X) = E(X^2) - E(X)^2$$

$E(X)^2$ is easy to compute, as it is simply the expected value raised to the power of 2. This means that $E(X^2)$ is what we need to concern ourselves with right now. If we write out the formula for $E(X^2)$ we get this:

$$E(X^2) = \sum_{r=1}^n \frac{r^2 \cdot \binom{R}{r} \binom{N-R}{n-r}}{\binom{N}{n}}$$

Using the law stated in eq. 7, we get:

$$E(X^2) = \frac{R}{\binom{N}{n}} \sum_{r=1}^n r \cdot \binom{R-1}{r-1} \binom{N-R}{n-r}$$

We again substitute $m = r - 1$ and $r = m + 1$:

$$E(X^2) = \frac{R}{\binom{N}{n}} \sum_{m=0}^{n-1} (m+1) \cdot \binom{R-1}{m} \binom{N-R}{n-1-m}$$

We can apply knowledge of sums to split up the sum at $m + 1$:

⁴ user6596353, and Michael Hardy. "Proving $Var(X) = E[X^2] - (E[X])^2$." *Mathematics Stack Exchange*, Stack Exchange, 18 July 2016, math.stackexchange.com/questions/1863562/proving-operatornamevarx-ex2-ex2.

$$E(X^2) = \frac{R}{\binom{N}{n}} \left(\sum_{m=0}^{n-1} m \cdot \binom{R-1}{m} \binom{N-R}{n-1-m} + \sum_{m=0}^{n-1} \binom{R-1}{m} \binom{N-R}{n-1-m} \right)$$

Again, we can use the manipulation we used in eq. 8 to have it look like this⁵:

$$E(X^2) = \frac{R \cdot \binom{N-1}{n-1}}{\binom{N}{n}} \left(\sum_{m=0}^{n-1} \frac{m \cdot \binom{R-1}{m} \binom{(N-1)-(R-1)}{(n-1)-m}}{\binom{N-1}{n-1}} + \sum_{m=0}^{n-1} \frac{\binom{R-1}{m} \binom{(N-1)-(R-1)}{(n-1)-m}}{\binom{N-1}{n-1}} \right)$$

Upon closer inspection, we can see that the first sum in the large parentheses will just equal the expected value of a hypergeometric distribution with population size

$N - 1$, population success count $R - 1$ and sample size $n - 1$. From our formula for the expected value, this turns into $(R - 1) \cdot \frac{n - 1}{N - 1}$. The second sum is familiar, and simply

equal to 1. Thus, this is what we have now:

$$E(X^2) = \frac{R \cdot \binom{N-1}{n-1}}{\binom{N}{n}} \left((R - 1) \cdot \frac{n - 1}{N - 1} + 1 \right)$$

Finally, we can use the same manipulation as earlier to get:

$$E(X^2) = \frac{Rn}{N} \left(\frac{(R - 1)(n - 1)}{N - 1} + 1 \right)$$

We know that $Var(X) = E(X^2) - E(X)^2$, so:

$$Var(X) = \frac{Rn}{N} \left(\frac{(R - 1)(n - 1)}{N - 1} + 1 \right) - E(X)^2 = \frac{Rn}{N} \left(\frac{(R - 1)(n - 1)}{N - 1} + 1 \right) - \frac{R^2 n^2}{N^2}$$

Factoring out $\frac{Rn}{N}$:

$$Var(X) = \frac{Rn}{N} \left(\frac{(R - 1)(n - 1)}{N - 1} + 1 - \frac{Rn}{N} \right)$$

⁵ Shahid, Iqbal. "Hypergeometric Distribution: Variance." *YouTube*, Google, Inc., 23 Apr. 2017, www.youtube.com/watch?v=yFIP8WVFrto.

$$\begin{aligned}
Var(X) &= \frac{Rn}{N} \left(\frac{Rn - R - n + 1}{N - 1} + 1 - \frac{Rn}{N} \right) \\
Var(X) &= \frac{Rn}{N} \left(\frac{NRn - NR - Nn + N}{N(N - 1)} + \frac{N^2 - N}{N(N - 1)} - \frac{NRn - Rn}{N(N - 1)} \right) \\
Var(X) &= \frac{Rn}{N^2(N - 1)} (NRn - NR - Nn + N + N^2 - N - NRn + Rn) \\
Var(X) &= \frac{Rn}{N^2(N - 1)} (N^2 - NR - Nn + Rn) \\
Var(X) &= \frac{Rn}{N^2(N - 1)} (N - n)(N - R) \tag{11}
\end{aligned}$$

This is the simplest version of the variance formula. Applying this to our marble example, and substituting values:

$$\sigma^2 = \frac{28 \cdot 20}{64^2(63)}(64 - 20)(64 - 28) \approx 3.44, \text{ and } \sigma \approx 1.85$$

Since this value for the standard deviation σ is the same as GeoGebra gives to us, we can be certain that we have done our calculations correctly.

7. Solving the opening problem

What is left now to do it apply all of this knowledge that we acquired to our opening problem, the *evil dares* game. We want to calculate the probability of a “loss”, meaning the probability of a player pulling less than or exactly 3 evil dares out of 7 from the collection of 50. In mathematical terms, this would mean to find $P(X \leq 3)$. The $\leq$ symbol signifies *cumulative probability*, which we must look at as well, but for now, let us define the distribution of random variable X : $X \sim HG(50, 25, 7)$. These terms come from our rules that I stated in the introduction. Let us, to start off, find the probability $P(X = 3)$, which we can then work with to find $P(X \leq 3)$. To find $P(X = 3)$, we use the formula:

$$P(X = 3) = \frac{\binom{25}{3} \binom{50 - 25}{7 - 3}}{\binom{50}{7}}$$

$$P(X = 3) = \frac{\binom{25}{3} \binom{25}{4}}{\binom{50}{7}}$$

$$P(X = 3) \approx 0.291$$

This demonstrates that the probability of having exactly 3 evil dares in your sample of 7 is around 0.291, or 29.1%. To find the cumulative probability $P(X \leq 3)$, we need to add all the probabilities below it, meaning

$P(X \leq 3) = P(X = 3) + P(X = 2) + P(X = 1) + P(X = 0)$. Using the same method, we find these values: $P(X = 3) \approx 0.291$, $P(X = 2) \approx 0.160$, $P(X = 1) \approx 0.0443$, and $P(X = 0) \approx 0.00481$. Now, we must sum these together. $0.291 + 0.160 + 0.0443 + 0.00481 \approx 0.500$. This confirms that $P(X \leq 3) = 0.5$. If we use the GDC (fig. 4 and 5) to find the cumulative probability, we also get the answer as a clean 0.5. What this means in terms of our problem is that the probability of a “loss” is exactly 1 in 2. We could therefore say the game is *fair*. The natural next step to this is to try and find the expected value and variance of this distribution.

The expected value can be seen as the average number of evil dares a player gets in their sample. To find it, we use our formula, eq. 9 and substitute values: $E(X) = 25 \cdot 7/50$, so $E(X) = 3.5$.

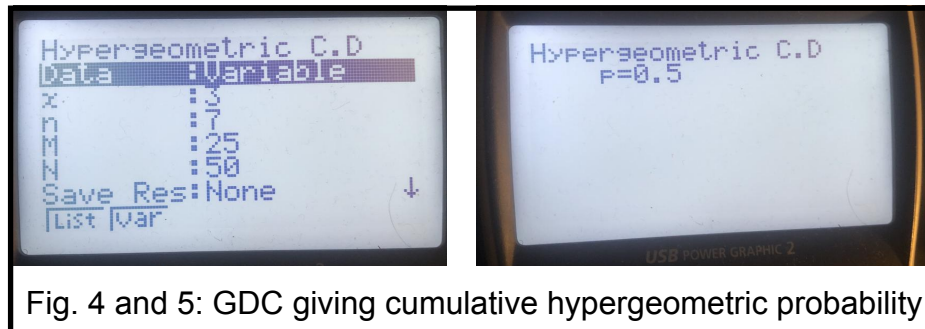


Fig. 4 and 5: GDC giving cumulative hypergeometric probability

Our mean is therefore 3.5, which makes sense as $r = 3$ and $r = 4$ both give the highest probability, and are equal. Since it is 3.5, we can round either up or down and receive a logical answer for our *expected value*. For further calculations, however, we would use 3.5. Finally, the variance of our opening problem. In the context of the real world, this would indicate how much the amount of evil dares in the sample vary as the game goes on. We again use our formula, eq. 11:

$$\begin{aligned} Var(X) &= \frac{25 \cdot 7}{50^2 \cdot 49} (50 - 7)(50 - 25) \\ &= \frac{175}{122,500} \cdot 43 \cdot 25 \\ Var(X) &= \sigma^2 \approx 1.54, \sigma \approx 1.24 \end{aligned}$$

This is a relatively low variance, so we would definitely be able to make the assumption that when playing the game multiple rounds, the amount of evil dares in the sample would mostly lie between 2 and 5. Fig. 6 is a bar graph which depicts the distribution of our evil dares game, which is a nice conclusion. Very notable are the values of 3 and 4 for r , which have equal probabilities, and give us our mean: $\frac{3 + 4}{2} = 3.5$.

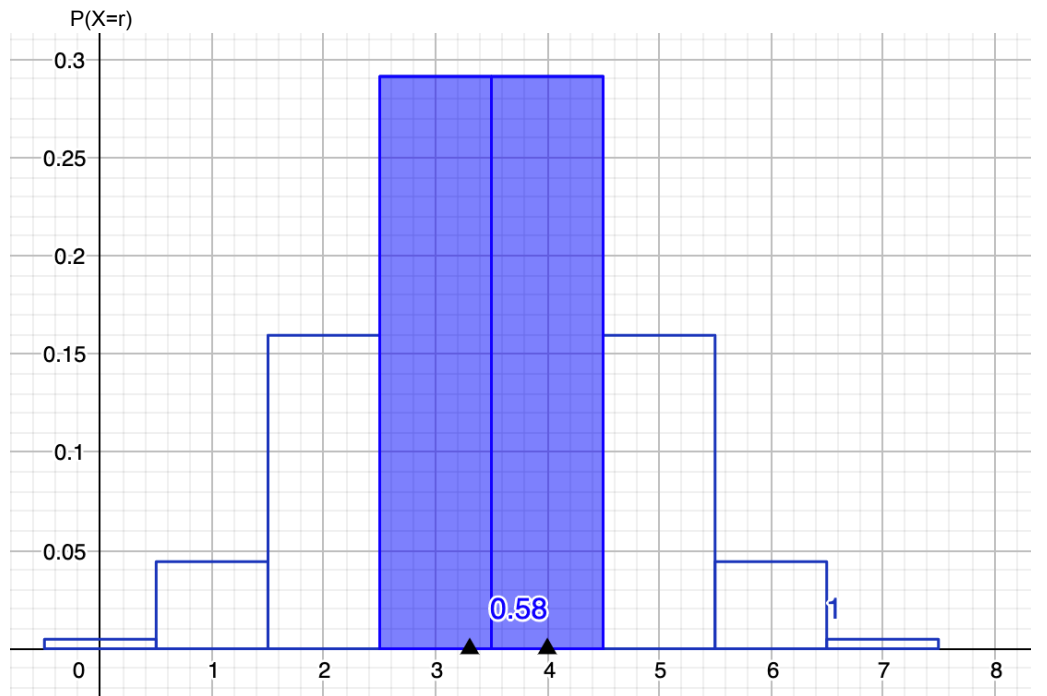


Fig. 6: Graph of $X \sim HG(50, 25, 7)$ (Own graph, Hohenwarter)

8. Conclusion

In this IA, we have looked in detail at the hypergeometric distribution and the different parameters used by it. Then we looked at the expected value and variance of such a distribution, and how we can derive the formulas for each. We also looked at how we can take a party game and find probabilities related to it. Having done a large number of calculations in this IA, it is fascinating to discover that the probability of winning or losing when playing the evil dares party game is 0.5, which is just equal to the probability of guessing the face right when tossing a coin. It is interesting to think about how accurate a probability model of a real life situation is. One would need to look at real life factors, such as that the distribution of the evil dares in the sample of 50 will probably not be distributed truly randomly, and the way people pick the dares. Also, rounding values would make the probabilities that we get not exactly that value. In conclusion, there obviously will be some “human” factor in the game, meaning that the probabilities shown in fig. 4 will probably not reflect the frequencies of a specific amount of evil dares in the sample, but the beautiful thing about math is that the *model* gives us an accurate enough understanding. Briefly, there are some areas we could further delve into to expand this IA. One of these is

cumulative frequency. While in our party game example, we did find a cumulative frequency, we did this by adding which is not very scalable if the amount of probabilities you need to add gets larger. A way around this is to find a *cumulative probability function*, which has the property that substituting a value of r gives $P(X \leq r)$. This would involve extra derivation and would not have transcended the scope of this IA. Also, areas like investigating further into the symmetry of probability mass functions, or perhaps finding $E(X)$ and $\text{Var}(X)$ formulas using probability generating functions.

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